RETAIL PERFORMANCE & MANAGEMENT REPORTING

Executive MBA - Excel Business Analytics Case Study

Your Role	Dataset	Primary Tool
Management Analyst	Retail Transaction Data	Microsoft Excel

The Case

You have joined the Business Analytics team of a multi-channel fashion retailer. The company sells through several online channels and serves both consumer and business customers. Management has access to transaction-level data, but the current file is primarily operational: it records individual sales lines without providing executives with a concise view of business performance.

Senior management wants an Excel-based reporting solution that transforms this raw data into useful management information. The report should show how the business is performing, identify the factors driving performance, highlight sales and fulfilment trends, and allow managers to analyse any selected reporting period without rebuilding the analysis.

> **Your task is to prepare the data, design the required performance measures, develop a dynamic management report, and use the resulting analysis to communicate meaningful business insights.**

Learning Objective

This case is designed to assess your ability to move from raw transactional data to management reporting. You are expected to demonstrate sound business logic, appropriate data preparation, accurate KPI design, effective reporting, and managerial interpretation. The objective is not simply to create charts or use a large number of Excel formulas.

1. Understanding the Dataset

The supplied dataset contains **31,047 transaction lines** representing retail activity during 2022. It contains **28,471 distinct orders** and **28,437 distinct customers**. This distinction is important: a transaction line, an order, a customer, a unit sold, and sales value are different business measures and should not be treated interchangeably.

Data Area	Examples of Available Fields
Transaction identifiers	Order ID, Customer ID
Customer profile	Gender, Age, Age Group
Time	Date
Fulfilment	Status: Delivered, Returned, Cancelled, Refunded
Sales channel	Amazon, Myntra, Flipkart, Ajio, Meesho, Nalli, Others
Product	SKU, Category, Size
Commercial measures	Quantity, Currency, Amount
Geography	Ship City, Ship State, Postal Code
Customer type	B2B indicator

Before beginning the analysis, examine the dataset yourself. Do not assume that every supplied column is immediately suitable for reporting.

2. Phase I - Data Preparation

Prepare the dataset for analysis and reporting. Review field names, data types, unnecessary fields, consistency of values, and any other issues that may affect reporting. The original raw data should be preserved so that your preparation process can be traced.

Time Dimensions

The Date field is the primary transformation requirement. Create appropriate reporting dimensions from it. At minimum, your prepared dataset should support analysis by **Year, Quarter, Month, Month Number, Week, Day, and Day Name**. Consider how these fields should be created and formatted so that chronological reporting behaves correctly.

Data Preparation Question

Which fields in the raw file should be retained, renamed, reformatted, transformed, or excluded before the data is used for management reporting? Be prepared to justify your decisions.

3. Phase II - Executive KPI Scorecard

Management first wants a concise scorecard showing the overall scale and quality of business performance. Develop KPI measures that correctly reflect the business definitions below. You must determine the appropriate Excel method and calculation logic.

KPI	Management Requirement
Total Customers	Number of distinct customers represented in the data
Total Orders	Number of distinct customer orders
Overall Sales	Total recorded sales value across all order statuses
Delivered Sales	Sales value associated with successfully delivered transactions
Returned Sales	Sales value associated with returned transactions
Cancelled Sales	Sales value associated with cancelled transactions
Refunded Sales	Sales value associated with refunded transactions
Delivered Sales %	Delivered sales as a percentage of overall sales
Non-Delivered Sales %	Combined returned, cancelled and refunded value relative to overall sales
Overall Quantity	Total quantity recorded
Delivered Quantity	Quantity successfully delivered
Returned Quantity	Quantity returned
Cancelled Quantity	Quantity cancelled
Refunded Quantity	Quantity refunded

Important: Management expects business measures, not row counts. Ensure that your definitions of orders, customers, sales and quantities are logically correct.

4. Phase III - Trend & Performance Reporting

Develop a time-based view of performance that allows management to understand how sales are changing. Your report should include monthly sales and an appropriate Month-on-Month comparison. It should also allow management to compare overall sales with delivered performance and quantity movement over time.

Choose suitable tables and visualisations. Your design should make trends, changes, unusual periods and potential performance issues easy to identify. Avoid adding a chart merely because a field exists.

5. Phase IV - Dynamic Date-Range Management Report

The most important requirement is a dynamic report controlled by two management inputs: **Start Date** and **End Date**. When these dates are changed, the analytical tables in this section must automatically recalculate for the selected period.

Reporting Area	Analysis Required for the Selected Date Range
Channel Performance	Orders, sales, delivered sales and contribution by sales channel
Customer Gender	Customers, orders, quantity and sales by gender
Product Category	Quantity, sales and delivered sales by category
Size Analysis	Demand by size, including quantity and delivered quantity
Order Status	Orders, quantity and sales by fulfilment status
B2B vs B2C	Comparison of orders, customers, quantity and sales
Geographical Performance	Performance by state and/or city using relevant business measures

The purpose of this section is not to produce several static summaries. A manager should be able to enter or select a different reporting period and immediately receive the corresponding analysis.

6. Management Questions

1. How much revenue is the business generating, and how much of that value is being successfully delivered?

2. What proportion of business value and quantity is affected by returns, cancellations and refunds?

3. How has sales performance changed month-on-month? Which periods show meaningful improvement or deterioration?

4. Which sales channels contribute most to orders, sales and delivered sales? Does the highest-volume channel also create the highest value?

5. What differences can be observed between male and female customers in customer count, orders, quantity and sales?

6. Which product categories and sizes generate the greatest demand? Are the highest-volume products also the highest-value products?

7. How significant is B2B activity compared with B2C activity in terms of customers, orders, quantity and sales?

8. Which geographical markets contribute most to performance, and are there locations where fulfilment performance requires attention?

9. When the Start Date and End Date are changed, which business drivers change most noticeably?

10. Based on the complete analysis, what are the most important issues or opportunities that senior management should investigate further?

7. Final Executive Challenge

Assume that your completed workbook will be reviewed by senior management. Conclude the case with **three significant business insights** and **three management actions or recommendations**. Each conclusion must be supported by evidence from your analysis rather than by assumption.

Required Submission

Workbook Component	Expected Content
1. Raw Data	Original dataset retained without destructive editing
2. Prepared Data	Cleaned reporting dataset with required date dimensions
3. KPI Report	Executive scorecard with correctly defined business KPIs
4. Trend Analysis	Monthly and Month-on-Month performance reporting
5. Dynamic Report	Start Date / End Date driven management analysis
6. Management Insights	Evidence-based findings and recommended actions

Assessment Focus

Your work will be assessed on the quality of data preparation, correctness of business definitions, accuracy of calculations, effectiveness of the dynamic reporting logic, clarity of presentation, and quality of managerial interpretation.

> **Remember: A strong management report is not the report with the most charts. It is the report that helps a decision-maker understand what happened, why it matters, and where attention is required.**

End of Case Study